

Ritesh Bhandari

Data Engineering & AI student ■ Student Assistant at CoRe, Turku UAS

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Profile

I am a Data Engineering and AI student finishing a B.Eng. at Turku University of Applied Sciences, focused on machine learning, local LLM tools, data pipelines and backend systems. I have contributed to real projects as a student assistant at CoRe and worked as a startup co-founder. For my bachelor's thesis, commissioned by the Renewing Car Recycling project at Turku UAS, I built a machine learning model that predicts used car part prices. I am self-driven and pick up new tools fast. I am co-authoring my first paper, on privacy in federated learning, and my long-term goal is to become a researcher.

Education

B.Eng., Information and Communications Technology

Aug 2023 – Jan 2027 (expected)

Turku University of Applied Sciences ■ Data Engineering and AI track

Turku

- Thesis: used car part price prediction, commissioned (see Projects).
- Key courses: Deep Learning, Data Analytics and Machine Learning, Big Data Engineering, Applications of AI.

Work Experience

Student Assistant

May 2026 – present

CoRe, Cognitive Technologies Research Group, Turku University of Applied Sciences

Turku

- Develop features and release tooling for an on-premises AI desktop application.
- Build internal tools that run local language models.
- Set up and maintain GPU servers and edge devices for running local AI models, and write setup guides for the team.
- Support the intern team with technical problems and project work.

AI Research Intern (practical training, 270 h)

Apr 2025 – Sep 2025

Cognitive Technologies Research Group, Turku University of Applied Sciences

Turku

- Co-shaped the Data Hub architecture, interoperability and API definition for THRIVE, a EUR 6.98M Horizon Europe project.
- Designed a normalized PostgreSQL schema with Alembic migrations, containerized the backend with Docker and test suites, and deployed it on CSC.
- Built an OCR pipeline (PaddleOCR, with Tesseract and EasyOCR fallbacks) for an industrial document-extraction prototype.

Co-founder, SisuSpeak (AI Finnish speaking practice)

Jun 2025 – Aug 2025

Startup Journey 2025 accelerator, Boost Turku

Turku

- Ran customer discovery and outreach, planned the speech pipeline of the MVP (speech-to-text, LLM, text-to-speech) and pitched the product at Demoday.

Lead Camp Instructor (2026), Camp Instructor (2024, 2025)

Summers 2024 – 2026

Lastenliitto, Helsinki district overnight camps

Finland

- Organized and led a 2026 overnight camp with a team of 5 instructors and 30+ children.

Waiter, part-time on weekends

ongoing

Restaurant

Projects and Tools Developed

Bachelor's thesis: used car part price prediction (commissioned)

github.com/Riteshbhandarii/DPPM

Python, scikit-learn, pandas, SHAP ■ 2026

- Commissioned by the Renewing Car Recycling project (Uudistuva autonkierrätys) at Turku UAS. Built a machine learning model that estimates the asking price of used car parts from marketplace listings and vehicle registry data.

- Built the whole pipeline myself, from scraping and cleaning to training, SHAP analysis and a working prototype.
- Designed a strict evaluation that keeps the same part out of both training and test data, compared the model with a simple pricing baseline, and validated it on newly collected live listings.

Data anonymization R&D project (team lead)

github.com/Riteshbhandarii/data-anonymization

Python ▪ Sep 2026 – present

- Lead and organize a six-person team building tooling that removes sensitive data from company documents, images, audio and CAD files before they are sent to public AI models.

Federated learning security benchmark (Innovation Project, team)

github.com/imaddde867/FL-Attack

Python, PyTorch ▪ 2025

- Measures how much private training data leaks from federated learning updates. I built an early gradient-inversion attack and the differential privacy and encryption defences.

AIS vessel analytics (course project, team lead, team of 4)

github.com/Riteshbhandarii/ais-analytics

Python, PostgreSQL, MQTT, Jupyter

- Designed the architecture and coordinated the team. Built live ingestion of Digitraffic AIS messages over MQTT into PostgreSQL, with ETL, position prediction, an API and a map dashboard.

Reinforcement learning self-driving car (course, ongoing)

github.com/Riteshbhandarii/rf-donkeycar

Python, PyTorch, stable-baselines3 ▪ 2026

- Training driving policies in the DonkeyCar simulator with behavioural cloning and PPO.

Raspberry Pi temperature sensing (Project Hatchery course, team of 7)

Raspberry Pi, Python, Telegram Bot API ▪ 2024

- Wired a temperature sensor to a Raspberry Pi and worked on the Python Telegram bot that reports it.

TEORIAT chess engine (personal project)

chess-engine-two.vercel.app

Python, PyTorch, PostgreSQL, FastAPI, React

- Trained only on my own Chess.com games so it plays like me, not like a perfect engine. PyTorch LSTM model, FastAPI backend and a playable web app.

NightBowl portfolio site (personal project)

nightbowl.live

Astro, three.js, TypeScript, Cloudflare

- My portfolio as an interactive 3D ramen stall, with a content editor and its own domain.

UmbraChat real-time chat app (personal project)

github.com/Riteshbhandarii/chat-systems

Django Channels, WebSockets, Redis, PostgreSQL

- Private and group chats, read receipts, friend requests and GDPR data controls, deployed on Railway.

Skills

Software and interfaces: Python, TypeScript, Electron, SolidJS, Streamlit, FastAPI, Django, HTML/CSS

Data and ML: PostgreSQL/TimescaleDB, SQLite, Redis, MQTT, Airflow, pandas, scikit-learn, PyTorch, XGBoost/CatBoost, SHAP, OCR, local LLM serving (vLLM, Ollama)

Hardware and edge: Raspberry Pi, Jetson AGX Orin setup (JetPack, CUDA), GPU inference servers

DevOps and tools: Git, GitHub Actions, Docker, Linux (Ubuntu), Windows, macOS, CSC (HPC)

Training

C++ Programming ▪ Azure Fundamentals

2026

AWS Academy Graduate training badges: Data Engineering, Cloud Foundations, Cloud Operations, Machine Learning for NLP

2024 – 2026

Languages and Military Service

Finnish: fluent ▪ English: fluent ▪ Nepali: native ▪ Swedish: intermediate (B1 course, autumn 2026)

Conscript service, Finnish Defence Forces; military police, reservist

Jan 2022 – Sep 2022